Quadratic Functions and Transformations

Concept

A quadratic graph is a parabola. Its equation form reveals different features.

Vertex form

$y = a(x - h)^2 + k$; vertex (h,k), symmetry $x = h$.

Opening / range

$a > 0$: $y \geq k$; $a < 0$: $y \leq k$, on the full real domain.

Other forms

$ax^2 + bx + c$ shows y-intercept c; $a(x - r)(x - s)$ shows zeros.

Vertex calculation

$h = -b/(2a)$; $k = f(h)$, or complete the square.

Example 1

Describe $y = -2(x - 3)^2 + 8$ and find its zeros.

Solution

1. Vertex $(3,8)$, axis $x = 3$; opens down; maximum 8.
2. Set $y = 0$: $(x - 3)^2 = 4 \rightarrow x = 1$ or 5.
3. Range $y \leq 8$; domain all real x.

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Example 2

Convert $y = x^2 - 6x + 5$ to vertex form.

Solution

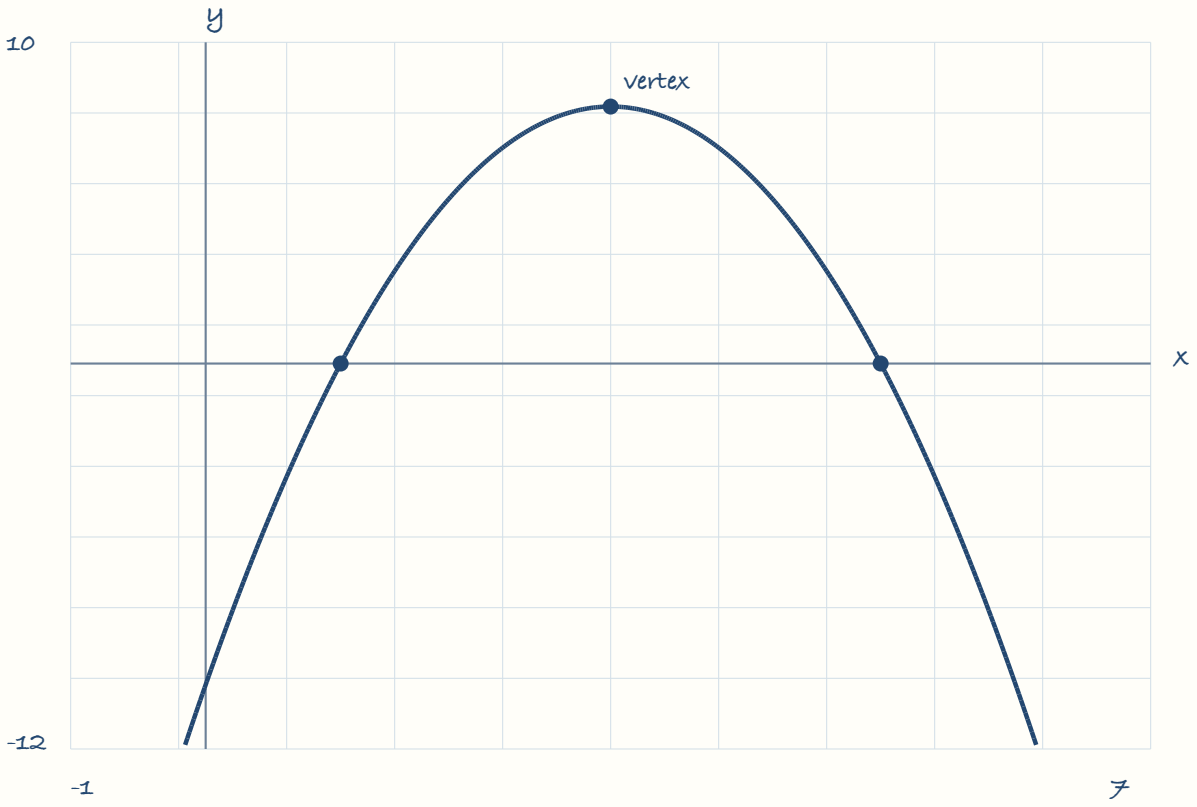
1. Complete square: $x^2 - 6x + 9 - 9 + 5$.

2. $y = (x - 3)^2 - 4$; vertex $(3,-4)$.

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Visual notes | read the graph

$y = -2(x - 3)^2 + 8$



The zeros 1 and 5 are equally far from the axis $x = 3$. The maximum output is 8.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. State vertex, axis and range of $y = 3(x + 1)^2 - 7$.

2. Convert $y = x^2 + 4x + 1$ to vertex form.

3. Find the y-intercept and zeros of $y = (x - 2)(x + 4)$.

4. A parabola has vertex (0,5) and passes through (2,-3). Find its equation.

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Practice answers | check after trying

1. Vertex $(-1,-7)$, axis $x = -1$, range $y \geq -7$.

2. $(x + 2)^2 - 3$.

3. y-intercept $(0,-8)$; zeros $x = 2$ and -4.

4. $-3 = 4a + 5$ gives $a = -2$, so $y = -2x^2 + 5$.